

# Image Processing

Dr: Reham Amin



# *Type of filters*

1. Weighted Average Filter.
  2. The median filter.
  3. laplacian filter
- 

# • Weighted Average Filter.

- Example 1: Apply weighted average filter to this example

|     |     |     |     |
|-----|-----|-----|-----|
| 10  | 20  | 30  | 40  |
| 50  | 60  | 70  | 80  |
| 90  | 100 | 110 | 120 |
| 130 | 140 | 150 | 160 |

**input image**  
**(without padding)**

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

**mask(Kernal)**

# solution

$$1^{\text{st}} \text{ pixel} = 1/16(1 \times 10 + 2 \times 20 + 1 \times 30 + 2 \times 50 + 4 \times 60 + 2 \times 70 + 1 \times 90 + 2 \times 100 + 1 \times 110) = 60$$

$$2^{\text{nd}} \text{ pixel} = 1/16(1 \times 20 + 2 \times 30 + 1 \times 40 + 2 \times 60 + 4 \times 70 + 2 \times 80 + 1 \times 100 + 2 \times 110 + 1 \times 120) = 70$$

⋮

⋮

$$4^{\text{th}} \text{ pixel} = 1/16(1 \times 60 + 2 \times 70 + 1 \times 80 + 2 \times 100 + 4 \times 110 + 2 \times 120 + 1 \times 140 + 2 \times 150 + 1 \times 160) = 110$$

**output image**

$$\begin{bmatrix} 60 & 70 \\ 100 & 110 \end{bmatrix}$$

**. Example 2: Apply weighted average filter to this example**

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 2 | 4 | 5 | 0 |
| 0 | 5 | 2 | 5 | 2 | 0 |
| 0 | 1 | 1 | 3 | 6 | 0 |
| 0 | 2 | 4 | 6 | 7 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 |

**input image**  
**(with one level padding)**

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

**mask(Kernal)**

# solution

$$1^{\text{st}} \text{ pixel} = 1/16(1 \times 4 + 2 \times 2 + 5 \times 2 + 2 \times 1) = 1$$

2<sup>nd</sup> pixel =  $1/16(1 \times 2 + 2 \times 4 + 4 \times 2 + 5 \times 1 + 2 \times 2 + 5 \times 1) = 2$

$$16^{\text{th}} \text{ pixel} = 1/16(3 \times 1 + 6 \times 2 + 6 \times 2 + 7 \times 4) = 3$$

**output image**

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 4 & 3 \\ 2 & 3 & 4 & 4 \\ 1 & 2 & 4 & 3 \end{bmatrix}$$

- **Example 3: Apply weighted average filter to this example**

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 15  | 25  | 35  | 45  | 55  |
| 65  | 75  | 85  | 95  | 105 |
| 115 | 125 | 135 | 145 | 155 |
| 165 | 175 | 185 | 195 | 205 |
| 215 | 225 | 235 | 245 | 255 |

**input image**  
**(without padding)**

$$\frac{1}{32} \begin{bmatrix} 2 & 4 & 2 \\ 4 & 8 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

**mask(Kernal)**

# solution

$$1^{\text{st}} \text{ pixel} = 1/32(15 \times 2 + 25 \times 4 + 35 \times 2 + 65 \times 4 + 75 \times 8 + 85 \times 4 + 15 \times 2 + 125 \times 4 + 135 \times 2) = 96$$

$$2^{\text{nd}} \text{ pixel} = 1/32(25 \times 2 + 35 \times 4 + 45 \times 2 + 75 \times 4 + 85 \times 8 + 95 \times 4 + 125 \times 2 + 135 \times 4 + 145 \times 2) = 109$$

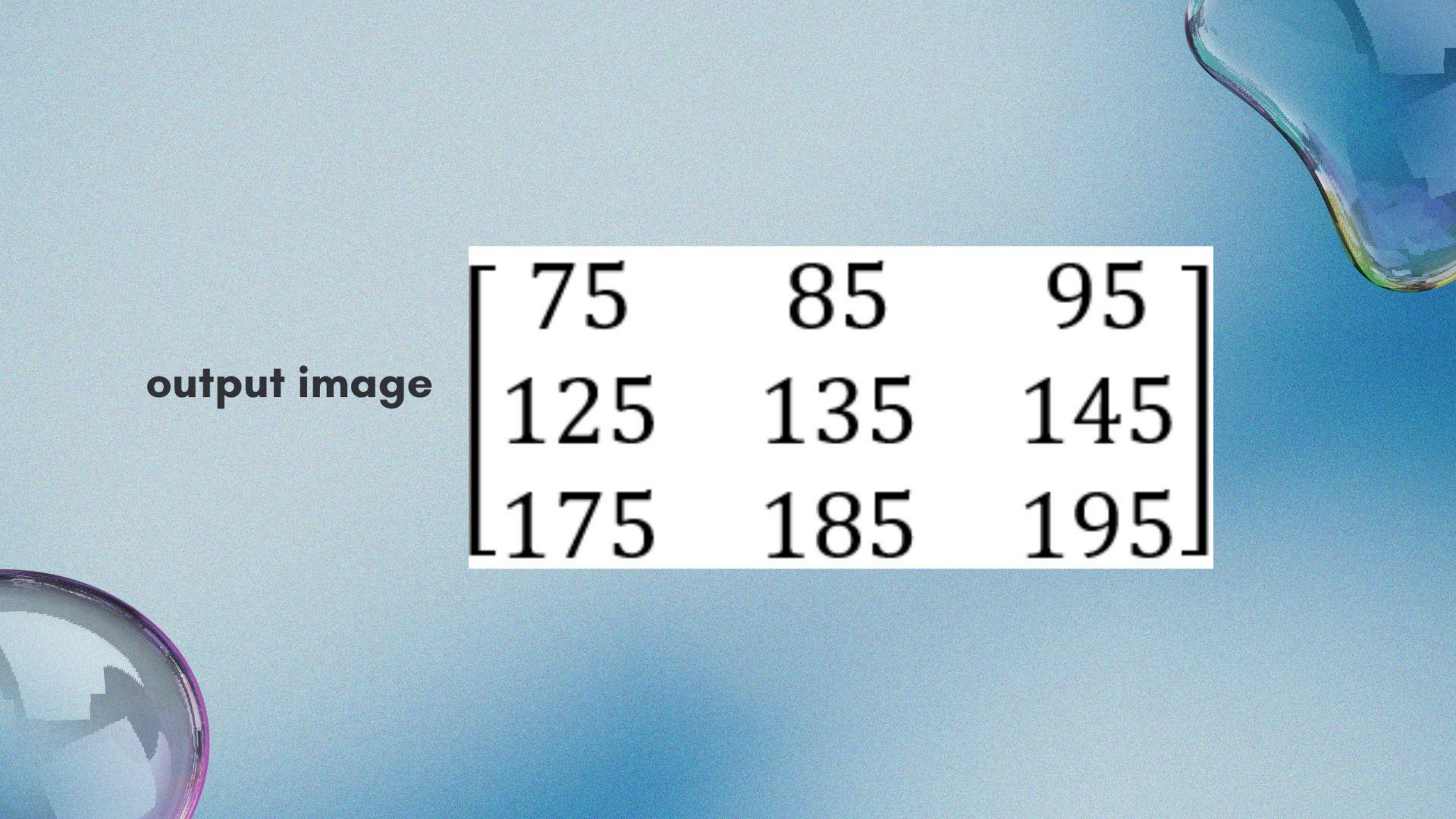
⋮

⋮

⋮

$$9^{\text{th}} \text{ pixel} = 1/32(135 \times 2 + 145 \times 4 + 155 \times 2 + 185 \times 4 + 195 \times 8 + 205 \times 4 + 215 \times 2 + 235 \times 4 + 245 \times 2) = 250$$

**output image**



|     |     |     |
|-----|-----|-----|
| 75  | 85  | 95  |
| 125 | 135 | 145 |
| 175 | 185 | 195 |

- The median filter.

- **Example 1 :Apply median filter to this example**

-> with window 3x3

|           |           |           |    |    |    |    |    |
|-----------|-----------|-----------|----|----|----|----|----|
| <b>20</b> | <b>20</b> | <b>20</b> | 30 | 40 | 40 | 40 | 40 |
| <b>20</b> | <b>20</b> | <b>20</b> | 30 | 40 | 40 | 40 | 40 |
| <b>20</b> | <b>20</b> | <b>20</b> | 30 | 40 | 40 | 40 | 40 |
| 30        | 30        | 30        | 30 | 30 | 30 | 30 | 30 |
| 40        | 40        | 40        | 30 | 20 | 20 | 20 | 20 |
| 40        | 40        | 40        | 30 | 20 | 50 | 20 | 20 |
| 40        | 40        | 40        | 30 | 20 | 40 | 20 | 20 |
| 40        | 40        | 40        | 30 | 20 | 20 | 20 | 20 |

**input image(without padding)**

# solution

1<sup>st</sup> pixel  $\gg$  20 20 20 20 20 20 20 20 20  $\gg$  *median* = 20

2<sup>nd</sup> pixel  $\gg$  20 20 20 20 20 20 30 30 30  $\gg$  *median* = 20

$\vdots$

$\vdots$

36<sup>th</sup> pixel  $\gg$  20 20 20 20 20 20 20 40 50  $\gg$  *median* = 20

**output image**

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 20 | 20 | 30 | 40 | 40 | 40 |
| 20 | 30 | 30 | 30 | 40 | 40 |
| 30 | 30 | 30 | 30 | 30 | 30 |
| 40 | 30 | 30 | 30 | 20 | 20 |
| 40 | 40 | 30 | 30 | 20 | 20 |
| 40 | 40 | 30 | 30 | 20 | 20 |

- **Example 2: Apply median filter to this example**

-> with window 3x3

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 10  | 20  | 30  | 40  | 50  |
| 60  | 70  | 80  | 90  | 100 |
| 110 | 120 | 130 | 140 | 150 |
| 160 | 170 | 180 | 190 | 200 |
| 210 | 220 | 230 | 240 | 250 |

**input image(without padding)**

# solution

1<sup>st</sup> pixel  $\gg$  10 20 30 60 70 80 110 120 130  $\gg$  median=70

2<sup>nd</sup> pixel  $\gg$  20 30 40 70 80 90 120 130 140  $\gg$  median=80

⋮ ⋮ ⋮

36<sup>th</sup> pixel  $\gg$  130 140 150 180 190 200 230 240  
250  $\gg$  median= 190

**output image**

|     |     |     |
|-----|-----|-----|
| 70  | 80  | 90  |
| 110 | 130 | 140 |
| 170 | 180 | 190 |

**. Example 3: Apply median filter to this example**

-> with window 2x2

|    |    |    |    |
|----|----|----|----|
| 12 | 15 | 18 | 20 |
| 21 | 25 | 30 | 35 |
| 40 | 45 | 50 | 55 |
| 60 | 65 | 70 | 75 |

**input image(without padding)**

# solution

1<sup>st</sup> pixel  $\gg$  12 15 21 25  $\gg$  median =  $(15+21)/2 = 18$

2<sup>nd</sup> pixel  $\gg$  15 18 25 30  $\gg$  median =  $(18+25)/2 = 22$   
:  
:  
:

9<sup>th</sup> pixel  $\gg$  50 55 70 75  $\gg$  median =  $(55+70)/2 = 63$

**output image**

|    |    |    |
|----|----|----|
| 18 | 33 | 53 |
| 22 | 38 | 58 |
| 24 | 43 | 63 |

- laplacian Filter.
- Example 1 :Apply laplacian filter to this example

|   |    |    |    |   |
|---|----|----|----|---|
| 0 | 0  | 0  | 0  | 0 |
| 0 | 10 | 10 | 10 | 0 |
| 0 | 10 | 20 | 10 | 0 |
| 0 | 10 | 10 | 10 | 0 |
| 0 | 0  | 0  | 0  | 0 |

input image  
(with one level padding)

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | 4  | -1 |
| 0  | -1 | 0  |

mask(Kernal)

# solution

$$1^{\text{st}} \text{ pixel} = (0 \times 0) + (-1 \times 0) + (0 \times 0) + (-1 \times 0) + (4 \times 10) + (-1 \times 10) + (0 \times 0) + (-1 \times 10) + (0 \times 20) = 20$$

$$2^{\text{nd}} \text{ pixel} = (0 \times 0) + (-1 \times 0) + (0 \times 0) + (-1 \times 10) + (4 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 20) + (0 \times 10) = 0$$

$$\begin{array}{ccc} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{array}$$
$$9^{\text{th}} \text{ pixel} = (10 \times -1) + (10 \times -1) + (10 \times 4) = 20$$

**output image**

$$\begin{bmatrix} 20 & 0 & 20 \\ 0 & 40 & 0 \\ 20 & 0 & 20 \end{bmatrix}$$

- **Example 2 :Apply laplacian filter to this example**

|    |    |    |    |    |
|----|----|----|----|----|
| 10 | 10 | 10 | 10 | 10 |
| 10 | 20 | 20 | 20 | 10 |
| 10 | 20 | 30 | 20 | 10 |
| 10 | 20 | 20 | 20 | 10 |
| 10 | 10 | 10 | 10 | 10 |

**input image**

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | 4  | -1 |
| 0  | -1 | 0  |

**mask(Kernal)**

# solution

$$\begin{aligned} 1^{\text{st}} \text{ pixel} = & (0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 10) + (4 \times 20) + \\ & (-1 \times 20) + (0 \times 10) + (-1 \times 20) + (0 \times 30) = 20 \end{aligned}$$

$$\begin{aligned} 2^{\text{nd}} \text{ pixel} = & (0 \times 10) + (-1 \times 10) + (0 \times 10) + (-1 \times 20) + (4 \times 20) + \\ & (-1 \times 20) + (0 \times 10) + (-1 \times 30) = 0 \end{aligned}$$

⋮

⋮

⋮

$$\begin{aligned} 9^{\text{th}} \text{ pixel} = & (0 \times 30) + (-1 \times 20) + (0 \times 10) + (-1 \times 20) + \\ & (4 \times 20) + (-1 \times 10) + (-1 \times 10) = 20 \end{aligned}$$

**output image**

$$\begin{bmatrix} 20 & 0 & 20 \\ 0 & 40 & 0 \\ 20 & 0 & 20 \end{bmatrix}$$

- **Example 3 :Apply laplacian filter to this example**

$$\begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & 5 \\ 5 & 5 & 5 \end{bmatrix}$$

**input image  
(without padding)**

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

**mask(Kernal)**

# solution

$$\begin{aligned} 1^{\text{st}} \text{ pixel} = & (1 \times 5) + (1 \times 5) + (1 \times 5) + (1 \times 5) + (-8 \times 10) + (1 \times 5) + \\ & (1 \times 5) + (1 \times 5) + (1 \times 5) = -40 \end{aligned}$$

**output image**

**$[-40]$**

# Our team

- *Ahmed Wael Mohammed*
- *Arwa Hatem Hamdy*
- *Asmaa Abdelrahman Badea*
- *Asmaa Ragab Abdelfattah*
- *Rania Salah Abdelsalam*
- *Mayada Gamal Abdelhamied*
- *Abdelrahman Ali Elsaid*
- *Ahmed Yasser*



Thank  
You